

NEMO

Next Generation Meta Operating System

D3 Integration Documentation

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List of Acronyms

Abbreviation / acronym	Description
AI	Artificial Intelligence
Dx.y	Deliverable number y belonging to WP x
EC	European Commission
EV	Electric Vehicle
HQ	Headquarter
IoT	Internet of Things
kWh	Kilowatt-hour
ML	Machine Learning
MVP	Minimum Viable Product
OSS	One Stop Shop
PV	photovoltaic
WP	Work Package

Executive Summary

Eros4NRG is a platform designed to monitor and analyse energy production and consumption, enabling more efficient decision-making through predictive analytics. It acts as a one-stop-shop (OSS) for energy stakeholders by integrating data from electric vehicles (EVs), EV charging stations, photovoltaic (PV) plants, and smart headquarters. The platform enhances trust and transparency in energy data, while addressing key challenges such as: (i) Unreliable data from IoT sources impacting machine learning operations; (ii) poor organization of static energy data leading to information loss; and (iii) the need for user-friendly AI/ML insights for non-technical stakeholders.

The primary difference compared to the last deliverable (D2.0 MVP Documentation) is the addition of the Zero-Trust Security and AI-Driven Predictive Analysis Modules. Eros4NRG's final integration includes six key modules:

1. *Data Gathering Module*: Continuously collects raw data from sources like EV fleets and PV systems, storing it in a MinIO data lake for efficient, scalable processing.
2. *Data Processing Module*: Transforms raw data into structured formats in PostgreSQL for querying, visualization, and AI/ML model training, ensuring data quality and reliability.
3. *Data Visualization Module*: Provides an intuitive dashboard for real-time and historical data insights, offering metrics like battery status, charging sessions, and energy consumption to support informed decision-making.
4. *Data Catalogue and API Module*: Offers an API interface for querying structured data, enabling users to retrieve and analyse both real-time and historical data easily.
5. *Zero-Trust Security Module*: User-based and resource-based authentication and authorization using keycloak integration. This is the module that integrates security provided by NEMO whereby NEMO provides the realm access and generates the token for authorization.
6. *AI-Driven Predictive Analysis*: AI Models integration to forecast energy production, energy consumption, optimize operations for PV, and EV battery health, and improve charging station usage.

This document also outlines the next steps required to further enhance the capability of the product and refine the forecasting:

1. *Stakeholders feedback loop*: Establish a continuous feedback mechanism to refine predictions and data processes based on stakeholder needs.
2. *Continuous AI model and data refinement*: Improve model performance by tuning parameters and enhancing data quality to better serve stakeholder requirements.
3. *Investigate enhancing predictive analysis with external factors*: Investigate integrating external data (e.g., weather conditions) to further improve predictive accuracy beyond historical database trends.

Integrated Eros4NRG establishes a solid foundation for monitoring and optimising energy systems through advanced data integration, processing, and visualization. With its robust architecture and modular approach, the platform is positioned to address the evolving needs of energy stakeholders by ensuring data accuracy, security, and usability. Moving forward, Eros4NRG will continue to expand its capabilities, refining predictive analysis for forecasting, improvement through stakeholder feedback, and enhancing the user interface for deeper insights and more intuitive decision-making.

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1 Introduction

Eros4NRG is a comprehensive platform for monitoring and analysing energy production and consumption. Through data-driven predictive analysis, Eros4NRG enables end users to make informed decisions about their energy operations, leading to more efficient business operations. This platform will also serve as a one-stop shop (OSS) for energy stakeholders. To realise this vision, we involved in the development process two main stakeholder groups:

- **EMOTION** is a company that manages a fleet of Electric Vehicles (EVs) and network of EV charging stations. It enabled Eros4NRG to access their EVs and EV charging stations databases to train the AI modules.
- **ASM Terni** is a company that owns a grid of photovoltaic plants (PVs) and distributes electricity to end users. ASM Terni provided Eros4NRG data on the PVs' power production and ASM Smart HQs power consumption.

In addition to energy data analytics, Eros4NRG aims to enhance data trust & transparency by analysing and securing IoT and edge data. Also, establishing cutting-edge data storage solutions that enhance semantic connections between dynamic information and metadata. Finally, Eros4NRG aims to enhance the Nemo Project Smart Energy Trial by acting as a native extension to it. To realise this vision, Eros4NRG will address the following key challenges:

- **Unreliable Energy IoT/Data Sources:** Addressing the impact of Inaccurate data in energy ecosystems (Buildings, PVs, EVs) on ML operations and training.
- **Lack of Organization and Management of Energy Data:** Tackling overlooked static data elements like energy entity attributes, environment, geolocation, and device fleets, leading to information loss and diminished analytics quality.
- **Involving Stakeholders in Cross-Value Chain Services:** Enhancing AI/ML model explainability and translating machine learning predictions into user-friendly language to make them comprehensible to non-technical stakeholders.
- **Deliver a one-stop-shop (OSS) for predictive energy analytics:** An OSS has the potential to be a centralised platform that offers all the necessary tools and resources for utilising predictive analytics in the energy sector. This platform will include features such as data collection, analysis, modelling, and forecasting, as well as tools for visualisation and interpretation of results.

1.1 Purpose of the document

This document details the final integration of the Eros4NRG platform. It outlines the key components and functionalities developed so far, with a focus on the following six main modules:

1. **Data Gathering Module:** Responsible for collecting raw data from various sources.
2. **Data Processing Module:** Managing data preprocessing, storage, and structuring for analysis.
3. **Data visualization Module:** Provides intuitive dashboards and reports for visualizing and interpreting the processed data.
4. **Data Catalogue and API Module:** Enables querying and managing the cleared data stored in the clean data bucket (the PostgreSQL database).
5. **Zero-Trust Security Module:** Integration of NEMO security module through keycloak realms. Enabling safe user- and resource-based authentication for API access
6. **AI-Driven Predictive Analysis:** AI Models integration for data forecasting using historical data

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This document also highlights the architecture of the platform and outlines the next steps for further development and improvements.

1.2 Relation to other project work

This is the third deliverable that presents the final integration of Eros4NRG following the second deliverable that detailed the MVP of Eros4NRG.

1.3 Structure of the document

This document is organized into two main chapters:

- **Chapter 2: Integrated Eros4NRG**

This chapter details the final integration of the six Eros4NRG modules.

- **Chapter 3: Next Steps**

This chapter outlines possible future steps to advance the Eros4NRG platform. These include the integration of external factors for improved AI-driven energy forecasting and continuous feedback for better addressing stakeholders' needs.

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2 Integrated Eros4NRG

Eros4NRG's final integration is guided by a set of clear functional and non-functional requirements, ensuring that the system operates efficiently across modern cloud-native infrastructures (such as MinIO, PostgreSQL, and Docker). Designed with a service-oriented architecture, the platform enables streamlined data collection, processing, visualization, and security.

The final Eros4NRG integration is composed of six core modules (Figure 1):

1. *Data Gathering Module*: Collects raw data from EV fleets, PV plants, and IoT devices, and stores it in a MinIO data lake.
2. *Data Processing Module*: Transforms and cleans the raw data, then transfers it into PostgreSQL for easier querying and analysis.
3. *Data Visualization Module*: Provides an interactive dashboard for stakeholders to monitor metrics such as battery status, energy production, and consumption trends.
4. *Data Catalogue (Eros4NRG API)*: Offers an API to retrieve cleaned and structured data, facilitating both real-time and historical data queries.
5. *Zero-Trust Security Module*: Integrates Keycloak-based authentication and authorization to protect data and ensure only authorized users gain access.
6. *AI-Driven Predictive Analysis Module*: Incorporates machine learning and deep learning models to forecast energy consumption, optimize battery health, and enhance overall operational efficiency.

By focusing on these six modules, Eros4NRG delivers a comprehensive solution for monitoring, analysing, and predicting energy usage, while also maintaining robust security and data integrity.

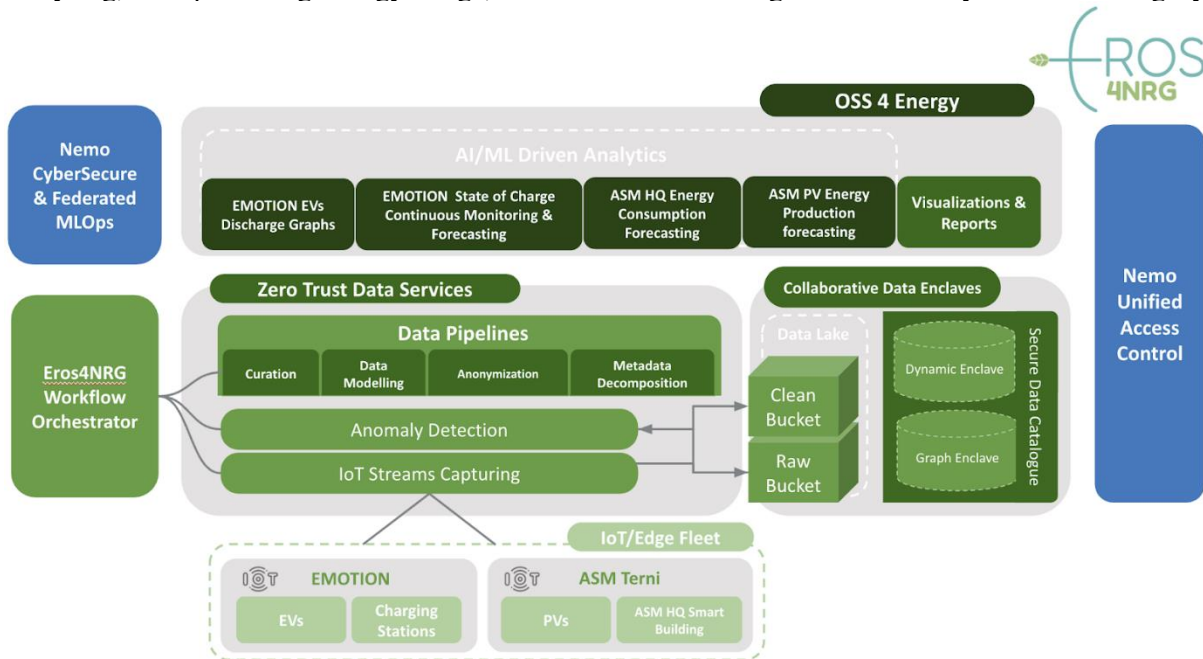


Figure 1: Eros4NRG's overall architecture

2.1 Data Gathering Module

Figure 2 illustrates the data gathering module. It is responsible for capturing raw data from EMOTION and ASM Terni systems via API calls. This continuous stream of data enables the platform to monitor and analyse energy consumption and production, as well as EV battery performance in (near) real time.

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The module starts by sending API requests to the EMOTION and ASM Terni systems. The collected data is fetched in chronological order, starting from the earliest available records to ensure the entire dataset is retrieved without missing any historical context. After the initial data collection, the system checks for new data entries every five minutes to maintain an up-to-date dataset.

The raw data is stored in a MinIO bucket acting as a data lake. MinIO is selected for its ability to efficiently manage large volumes of unstructured data, offering high scalability and rapid retrieval. This data is crucial for feeding downstream processes, including data preprocessing and anomaly detection. The MinIO architecture allows Eros4NRG to store data as soon as it is received, providing a streamlined approach to handling large data volumes from distributed sources such as PV plants and EV charging stations.

The data gathering module is further enhanced by using threads to parallelise the processes, increasing performance and reducing the time it takes to gather the data into MinIO. After the data gathering processes has finished running, an extra process then initiates a cleanup process that ensures there are no duplicates in the Postgres database to enhance data quality

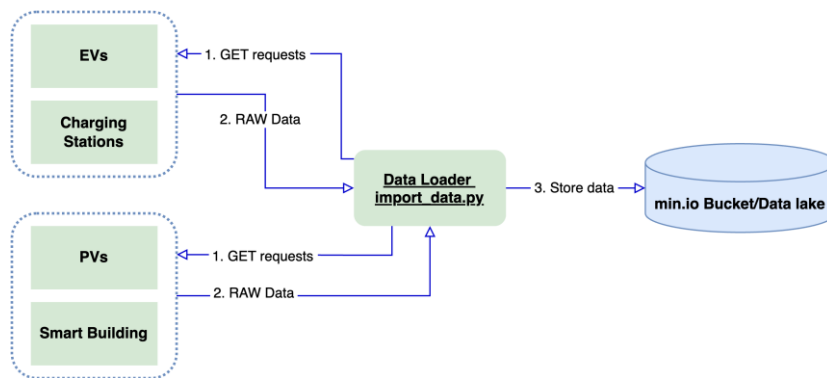


Figure 2: The data gathering module

2.2 Data Processing Module

The Data Processing Module is central to transforming the raw data into a structured format suitable for further analysis and querying. As illustrated in Figure 3, this module extracts the stored data from the MinIO bucket, applies preprocessing steps, and stores it in a relational database for efficient access.

The module starts by reading data from the MinIO bucket. It decomposes the data by separating metadata from actual data points. The processed data is then formatted as a DataFrame and stored in a PostgreSQL database. PostgreSQL, being a highly scalable and reliable relational database system, is used to facilitate easier query logic, especially for data visualization and data cataloging. This allows users to perform complex queries and retrieve specific data insights efficiently. Once data has been successfully transferred to PostgreSQL, it is deleted from MinIO to avoid data duplication and ensure storage optimization. This module not only enables the transformation of raw data into structured information but also lays the groundwork for advanced querying and AI/ML model training.

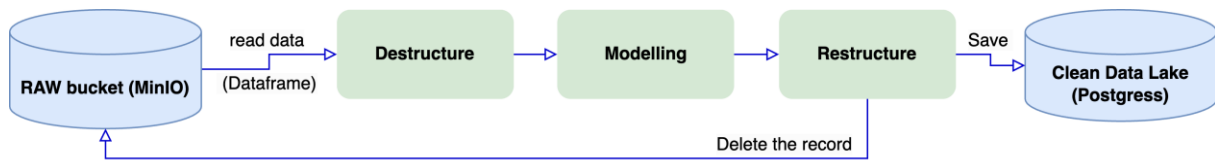


Figure 3: The data processing module

2.3 Data Visualization Module

The Eros4NRG dashboard, as shown in Figure 4, provides an intuitive interface for visualizing data stored in the PostgreSQL database. Users can select specific data providers and zoom in on the data of interest. For example, Figure 4 depicts data related to a vehicle from the EMOTION car fleet, identified as car ID 7. The displayed panels offer key metrics, such as the number of charges, battery percentage, charging status, and total kilometres driven. These visual insights help users understand trends, performance, and the overall condition of the vehicle fleet.

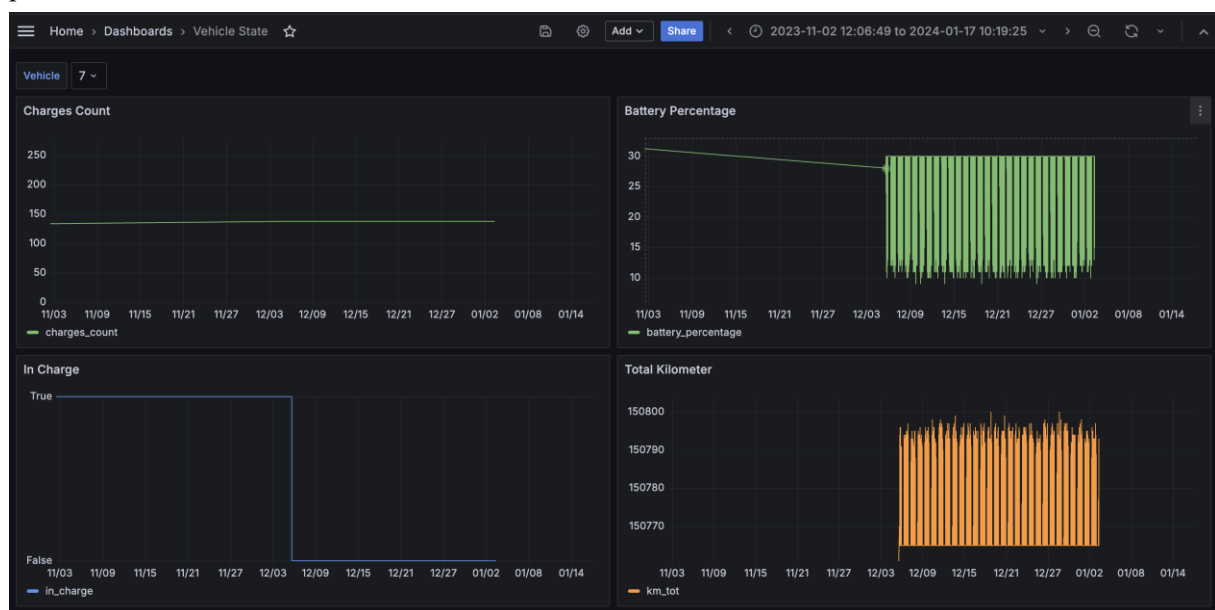


Figure 4: Eros4NRG Dashboard

The dashboard also allows for more granular data investigation over specific time frames. For instance, Figure 5 showcases data from vehicle ID 3, displaying battery percentage, charge amount, and total kilometres travelled. The graph reveals an anomaly in the kWh charged—it remains stagnant despite the vehicle covering more kilometres recently. This inconsistency, especially when the vehicle was charged at a low battery level, suggests potential issues with the sensor capturing the total kWh charged. This information is valuable for energy stakeholders, who can further investigate sensor accuracy and energy efficiency.



Figure 5: Information regarding vehicle 3

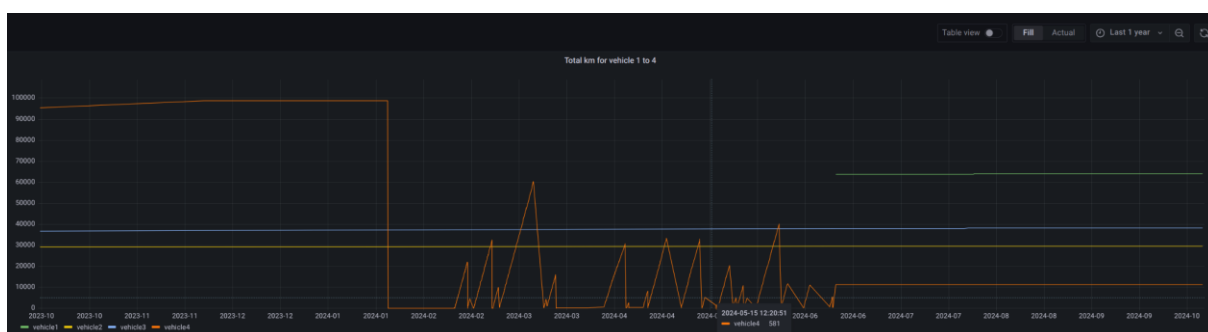


Figure 6 illustrates the total kilometres recorded between charges for vehicles 1, 2, 3, and 4. The graph indicates relatively consistent usage patterns for vehicles 1, 2, and 3. However, vehicle 4 exhibits significant variability in its usage, particularly from January to July 2024. This noticeable change invites energy stakeholders to investigate events in December 2023, as the vehicle showed higher mileage before that period.

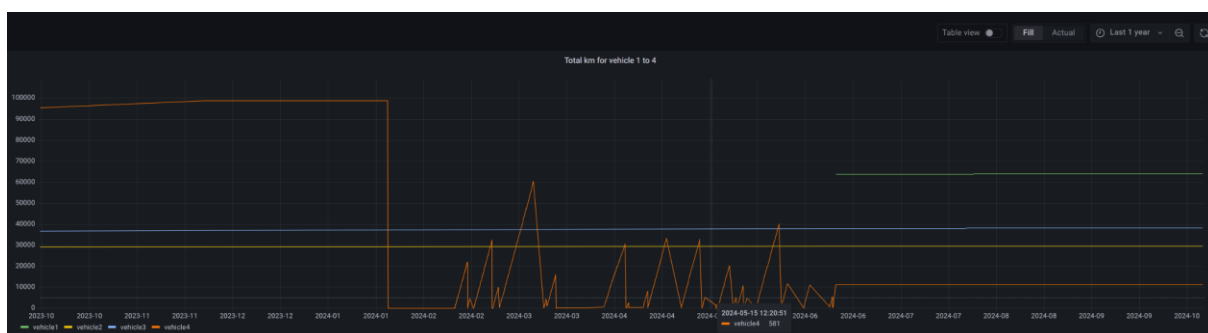


Figure 6: total kilometer for vehicle 1 to 4

Figure 7 presents a comparison of battery percentages recorded between charges for vehicles 1, 3, 8, and 9. The graph highlights that vehicles 1 and 3 were infrequently used from October to December 2023. In contrast, vehicle 8 showed consistent usage, as indicated by the progressively lower battery percentages with each charge. However, a noticeable shift occurred in June 2024, when vehicles 1 and 3 experienced a sharp drop in battery percentage, suggesting significant usage during that period. By July 2024, this trend extended to all four vehicles, all of which experienced substantial declines in battery percentage, continuing up to recent times.

Additionally, the data shows an anomaly for vehicle 8 in October 2023, where the recorded battery percentage spiked to an abnormal value of approximately 2264%, far exceeding the 100% limit. This

anomaly may be due to a sensor error or a misrecorded value. Energy stakeholders can use this insight to recommend a maintenance check for vehicle 8 to address potential sensor issues.

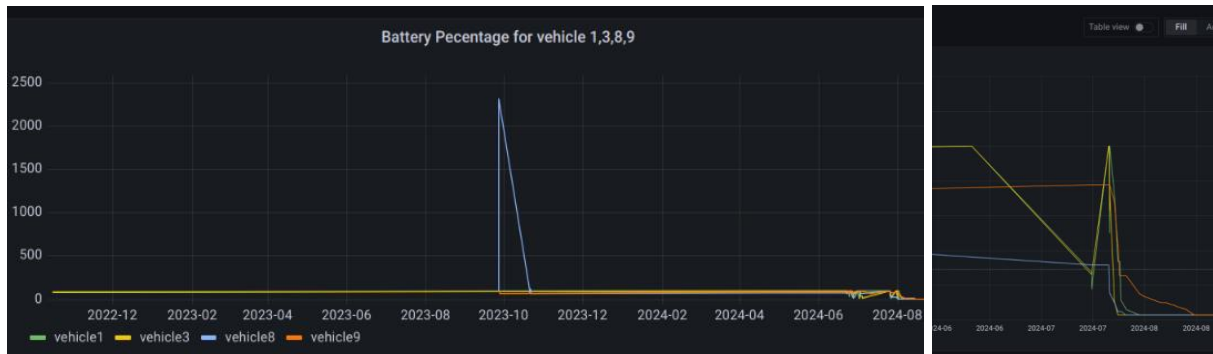


Figure 7: Battery percentage for vehicle 1,3,8,9

Figure 8 is the formula with which the calculation of the building consumption and energy production of the photovoltaic is done. These formulas are provided directly by ASM Terni and is processed automatically as soon as the data is stored in MinIO afterwards the calculation is then saved in the Postgres database.

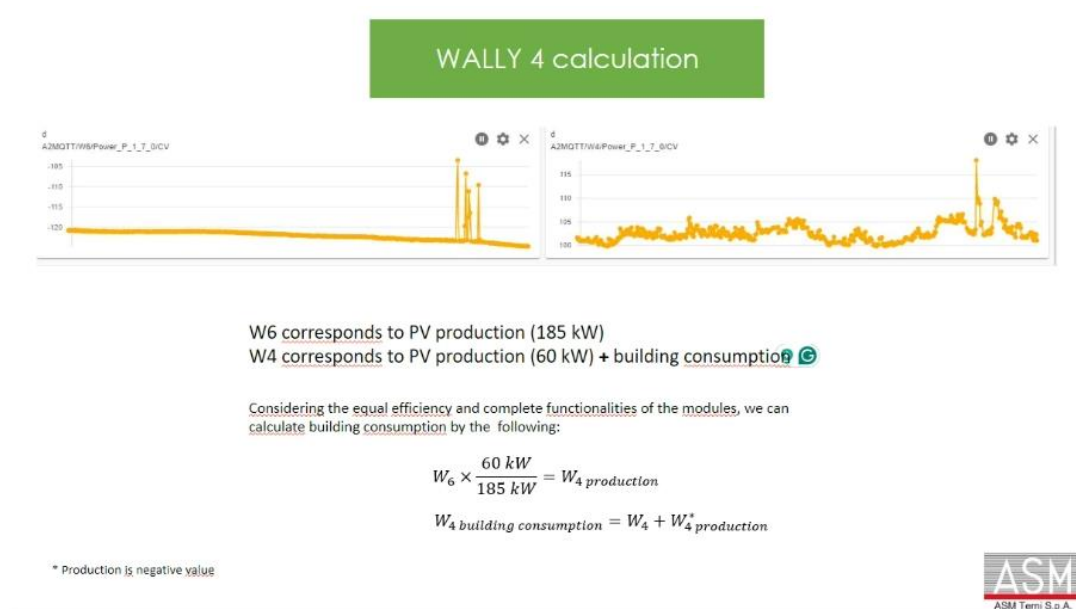


Figure 8: Formula for WALLY 4 calculations

Figure 9 presents the historical context of the calculated building consumption provided by the PV devices. The calculation is sampled on a fixed time interval and calculating the average value of the sampling, currently the sampling is done in 5 seconds interval.



Figure 9: WALLY 4 Building consumption

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Figure 10 provides the historical context of the calculated energy production from the PV devices. The sampling and calculation are done the same as the building consumption by using the average of 5 seconds interval



Figure 10: WALLY 4 energy production

2.4 Data Catalogue (Eros4NRG API)

The Data Catalogue and API Module (Figure 11) is an access point created using FastAPI to enable users to query the cleaned data stored in the Postgres database. It creates a request into the metadata table asking for specific information the user needs which then queries the history table that then returns the information asked by the user. Alternatively, it can also provide direct information from the database if the user already has the history ID that they want presented. Currently there are 6 endpoints, 2 of which are for accessing the metadata and 4 for directly accessing history information.

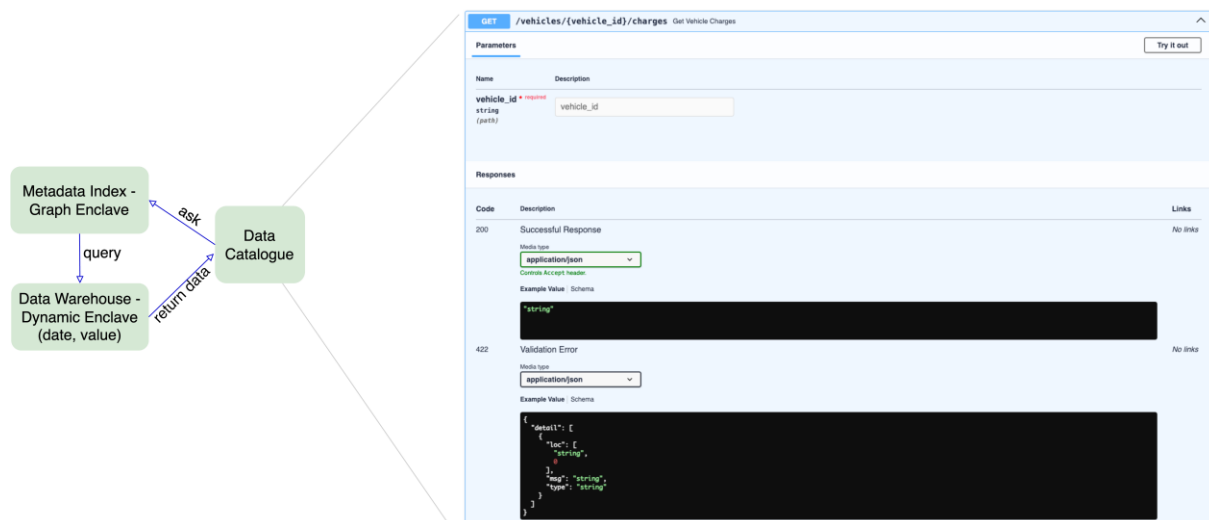


Figure 81: Data Catalogue

See Annexes for detailed information on the API calls and endpoints.

2.5 Zero-Trust Security Module

To ensure that our API comply with security standards and protect shareholder data, Eros4NRG integrates its authentication and authorization using Keycloak. This ensures a role-based and resource-based access control that will safeguard sensitive energy and IoT data from unauthorized access. This module integrates seamlessly with NEMO provided security protocol. It uses the realm provided directly by NEMO and all the users credentials can be managed from within the realm. After authentication the module allows access to the data resources provided in the Data Catalogue and API Module.

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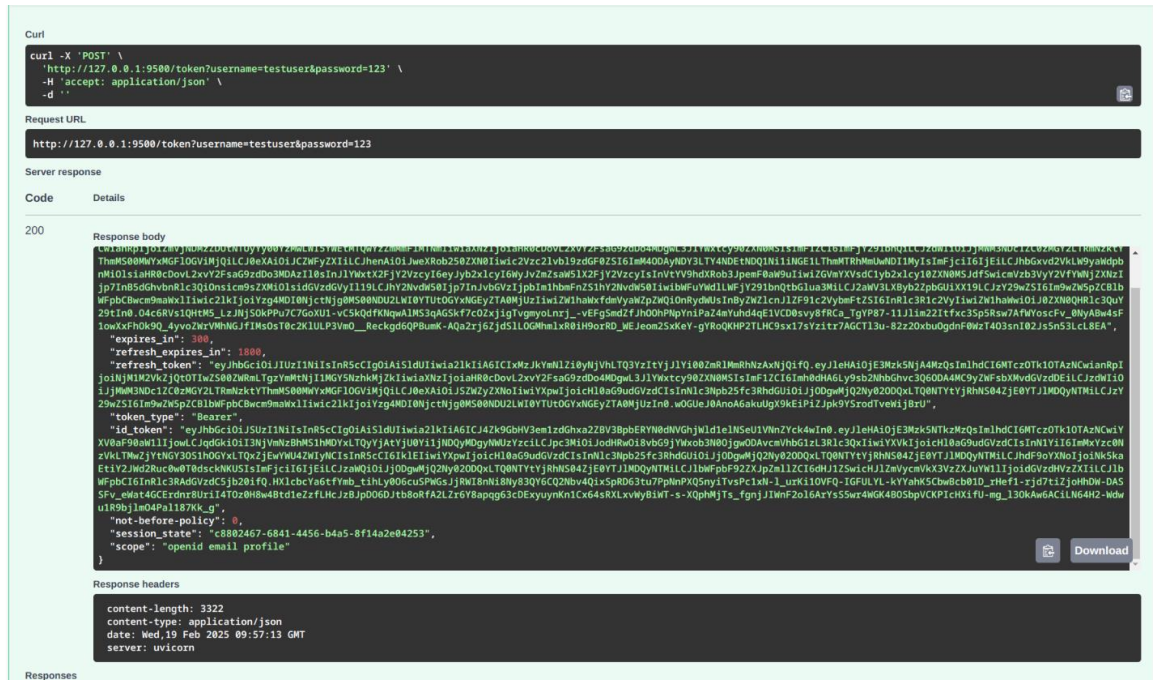


Figure 12: Zero-Trust Security

2.6 AI-Driven Predictive Analysis Module

The AI-Driven Predictive Analysis Module delivers insights by forecasting future values using historical data. After evaluating various algorithms—CatBoost, XGBoost, LightGBM, RandomForest, DecisionTree, and GradientBoost—it was found that CatBoost consistently produces the most accurate 12-hour-ahead forecasts for building consumption. In Figure 13, the green line depicts historical building consumption, the yellow line shows energy production, and the blue line represents CatBoost’s predictions. Although CatBoost performs well, purely machine-learning-based methods occasionally “flatline” due to cascading errors in the forecasting process. The forecasting is done at 6 a.m. the following day as per the request of ASM Terni.

To address these limitations, the next step involves refining deep learning models that capture more complex data patterns. Figure 14 highlights results from Prophet, a deep learning approach that analyses around a week’s worth of data to predict the following day’s consumption. Compared to other deep learning models tested, Prophet showcased the best performance, underscoring its potential for future enhancements in building consumption forecasting.



Figure 13: Building Consumption forecast using Catboost machine learning model

In Figure 14 we see the forecast result after using the Prophet deep learning model for the building consumption. The model takes in data from roughly a week into account and predicts the building

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consumption for the next day. The Prophet model performed the best out of the other deep learning models tested.



Figure 14: Building Consumption forecast using Facebook Prophet deep learning model

2.7 User manual and relevant libraries

To run the data gathering, processing, and visualization module, follow the following commands.

1. Clone the repository

```
Git clone https://github.com/Eros4NRG/Eros4NRG
```

2. Move inside the Eros4NRG folder

```
cd Eros4NRG
```

3. run the following docker-compose command in the terminal

```
docker-compose -f <<the .yaml filename>> up -d -build
```

4. Docker command to access individual services:

```
docker exec -it <<service name>> bash
```

5. Docker command to access log data for each service:

```
docker logs <<service name>>
```

6. Docker compose command to shut down the project

```
docker-compose -f <<the .yaml filename>> down -v
```

3 Next Steps

Building on the final integration, Eros4NRG will undergo several enhancements and additions to maximise its potential and improve user experience. The next steps in development include:

1. **Stakeholders feedback loop:** For continuous improvement and better integration, it is critical for there to be a feedback loop from the stakeholders to refine the predictions and data processing to further accommodate the needs and challenges of the stakeholders
2. **Continuous AI model and data refinement:** The AI models must be refined further by changing the parameters and improving data quality to fit the needs of the stakeholders
3. **Investigate enhancing predictive analysis with external factors:** Currently the predictive analysis is conducting the forecast using historical data based stored in the database. After receiving some feedback from stakeholders, it was suggested that further enhancement of the predictions could be made using data from external factors such as weather conditions. The possibility of these refinements and improvements needs to be investigated.

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4 Conclusions

This document has presented the final integration of the Eros4NRG platform, detailing the key components required for monitoring and analysing energy production and consumption. By addressing issues such as unreliable IoT data sources, ineffective energy data management, and the need for clearer AI/ML insights, Eros4NRG provides a solid starting point for comprehensive energy analytics.

Nevertheless, ongoing refinements are crucial to further enhance data quality and forecasting accuracy, particularly when integrating external factors (e.g., weather data) for predictive analysis. All six core modules—Data Gathering, Data Processing, Data Visualization, Data Catalogue, Zero-Trust Security, and AI Predictive Analysis—together establish a robust foundation for Eros4NRG. These modules will drive future deliverables focused on expanding predictive analytics capabilities and improving user interaction with energy data.

Looking ahead, the next steps involve implementing continuous stakeholder feedback loops and refining AI models, fully aligned with the overall project roadmap. These efforts will ensure that Eros4NRG evolves to meet both technical and operational demands, delivering more actionable insights and supporting informed decision-making across diverse energy ecosystems.

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5 Annexes

5.1 API Documentation

Eros4NRG provides an API module (based on FastAPI) that allows stakeholders to retrieve information directly from the Postgres database in a json format. Currently there are 3 endpoints that are provided

Metadata information regarding vehicles	
Description	List out available vehicle IDs and provides extracted metadata information regarding specific vehicles which is a list of charge history IDs
HTTP Method	GET
Endpoint	1. /vehicle 2. /vehicle=\${vehicle_id}
Parameters	Vehicle_id = the specific vehicle ID that the stakeholders would like metadata information from
Output	1. [1,2,3,4,5,6...] 2. [657841,324652, 626743,...]

Table 9: Vehicle Metadata endpoints

Information regarding vehicle charge history	
Description	Provides detailed information regarding specific vehicles charge history
HTTP Method	GET
Endpoint	/charges=\${charge_id}
Parameters	charge_id = the specific charge history ID that the stakeholders would like more detailed information from
Output	[Id Timestamp Charge_count Kwh_charged ...]

Table 10: Vehicle Charge History endpoint

Information regarding building consumption	
Description	Provides WALLY 4 calculated value of the building energy consumption
HTTP Method	GET
Endpoint	/building_consumption
Parameters	1. Start_date = the starting date you want the data from 2. End_date = the end date you want the data from
Output	[59.75,53.56,57.88,...]

Table 12: Building consumption history

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Information regarding energy production	
Description	Provides WALLY 4 calculated value of the PV energy production
HTTP Method	GET
Endpoint	/energy_production
Parameters	3. Start_date = the starting date you want the data from 4. End_date = the end date you want the data from
Output	[0.995,-13.12,-27.56,...]

Table 13: Energy production history

5.2 Eros4NRG Code Source

Eros4NRG's code is hosted publicly on GitHub and can be accessed via <https://github.com/Eros4NRG/Eros4NRG>